

The $d = 4$ fiber of the graded order law: a reduction to Schur support, and the obstruction

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Abstract

Let $G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$ be the graded branch-exponent generating function, where $s(T) = \sum_{i \in \text{Des}(T)} w_i$ with the parity weights $w_i = 2i - 1$ if $n - i$ is odd and $w_i = 0$ otherwise. At the cube root of unity the fiber value $G_\lambda(\zeta_3)$ vanishes exactly on the two 3-cores $(3, 1), (2, 1, 1)$, by a Schur-positivity argument [Clio 2026-06-02]. At the *fourth* root of unity $\zeta_4 = i$ the value is genuinely complex and that argument cannot transplant. We prove instead a structural *reduction*: because the active descent set is an arithmetic progression of common difference 2, at ζ_4 *all* of its factors in the master identity coincide, so that for $n = 2m$ even

$$G_\lambda(i) = \langle s_\lambda, \psi^m \rangle, \quad \psi := h_2 + i e_2 = s_2 + i s_{11},$$

a single pure power, with the odd- n analogue $G_\lambda(i) = \langle s_\lambda, h_1 \bar{\psi}^m \rangle$. Equivalently $G_\lambda(i) = P_\lambda(i)$ for the nonnegative-integer, conjugation-reciprocal polynomial $P_\lambda(t) = \langle s_\lambda, (h_2 + t e_2)^m \rangle$, so $G_\lambda(i) = 0 \iff (t^2 + 1) \mid P_\lambda(t) \iff \lambda$ lies outside the (complex) Schur support of ψ^m . We then prove two complete results. (i) *A parity criterion*: in $\mathbb{Z}[i]$ modulo the prime $1 + i$, $G_\lambda(i) \equiv i^m f^\lambda$, so f^λ odd implies $G_\lambda(i) \neq 0$, with the refinement $\text{Re } G_\lambda(i) \equiv \frac{1}{2}(f^\lambda + \chi^\lambda(2^m))$, $\text{Im } G_\lambda(i) \equiv \frac{1}{2}(f^\lambda - \chi^\lambda(2^m)) \pmod{2}$ and explicit mod-4 necessary conditions for a vanisher. (ii) *Exact values* from a weight- $\{1, i, 1 + i\}$ 2-strip recursion: $G_{(n)} = 1$, $G_{(1^n)} = i^m$, $G_{(2M-1, 1)} = (M - 1) + Mi$, and

$$\boxed{G_{(2m-2, 2)}(i) = m(m - 2)i},$$

the last vanishing *exactly* at $m = 2$, so within two-row shapes $(2, 2)$ is provably the unique vanisher. The general case reduces to full support of ψ^m (for $m \geq 3$; for $m = 2$ the support omits exactly $(2, 2)$, a one-line expansion). We isolate why this is hard – ψ^m has genuinely complex coefficients, the Pieri recursion admits cancellation, and the Schur-positivity that closed $d = 3$ is absent – and give two precise routes (a $(1 + i)$ -adic monovariant; a Hermitian bound from $\psi^m \bar{\psi}^m = \Xi^m$), each with its remaining obstruction.

1 Setup

Throughout $\lambda \vdash n$, and for $T \in \text{SYT}(\lambda)$, $\text{Des}(T) = \{i \in [n - 1] : i + 1 \text{ lies in a strictly lower row than } i\}$. Fix

$$A = \{k \in [n - 1] : n - k \text{ odd}\}, \quad w_k = (2k - 1) [k \in A], \quad s(T) = \sum_{i \in \text{Des}(T)} w_i,$$

and set $m = |A| = \lfloor n/2 \rfloor$, $e = n \bmod 2$. We work in the ring Λ of symmetric functions with the Hall inner product, for which $\langle s_\lambda, h_\mu \rangle = K_{\lambda\mu}$ and $\langle s_\lambda, p_\mu \rangle = \chi^\lambda(\mu)$. We use $h_2 = \frac{1}{2}(p_1^2 + p_2)$, $e_2 = \frac{1}{2}(p_1^2 - p_2)$, so $h_2 + e_2 = p_1^2$ and $h_2 - e_2 = p_2$. For $f \in \Lambda$ write $\text{supp}_{\mathbb{C}}(f) = \{\mu : \langle s_\mu, f \rangle \neq 0\}$.

We recall the master identity, proved (for any ζ) in the ζ_3 note.

Proposition 1 (Master identity). *For any $\zeta \in \mathbb{C}$, $G_\lambda(\zeta) = \left\langle s_\lambda, h_1^e \prod_{k \in A} (h_2 + e_2 \zeta^{2k-1}) \right\rangle$.*

2 The reduction: a single pure power at ζ_4

This is the structural phenomenon that distinguishes $d = 4$ from general d .

Lemma 1 (Coincidence of factors). *At $\zeta = i$, for every $k \in A$ the factor $h_2 + e_2 i^{2k-1}$ equals $h_2 + i e_2$ if n is even, and $h_2 - i e_2$ if n is odd. In particular all $|A| = m$ factors are equal.*

Proof. Since $2k - 1$ is odd, $i^{2k-1} = i$ if $2k - 1 \equiv 1 \pmod{4}$, i.e. k odd, and $i^{2k-1} = i^3 = -i$ if k even. The active set A is an arithmetic progression of common difference 2: $A = \{1, 3, \dots, n-1\}$ when n is even (all k odd) and $A = \{2, 4, \dots, n-1\}$ when n is odd (all k even). Hence within A the parity of k is constant, and i^{2k-1} takes a single value: i for n even, $-i$ for n odd. $\square$

Theorem 1 (Reduction). *Write $\psi = h_2 + i e_2$ and $\bar{\psi} = h_2 - i e_2$. Then*

$$n = 2m \text{ even: } G_\lambda(i) = \langle s_\lambda, \psi^m \rangle, \quad n = 2m + 1 \text{ odd: } G_\lambda(i) = \langle s_\lambda, h_1 \bar{\psi}^m \rangle.$$

Consequently, for n even, $G_\lambda(i) = 0$ if and only if $\lambda \notin \text{supp}_{\mathbb{C}}(\psi^m)$.

Proof. Immediate from Proposition 1 and Lemma 1: the product $\prod_{k \in A} (h_2 + e_2 i^{2k-1})$ collapses to ψ^m (resp. $\bar{\psi}^m$), and $h_1^e = 1$ (resp. h_1). The support statement is the definition of $\langle s_\lambda, \psi^m \rangle \neq 0$. $\square$

Remark 1. This is exactly the point where the ζ_3 proof's strategy breaks. At ζ_3 the factors split into a conjugate pair $\phi, \bar{\phi}$ whose product $\Theta = \phi \bar{\phi} = s_4 + 2s_{22} + s_{14}$ is *Schur-positive*, turning the fiber value into a manifestly nonnegative integer. At ζ_4 with n even all factors coincide as ψ ; there is no conjugate partner inside the product, so ψ^m has genuinely complex coefficients and no positivity is available. The value $G_\lambda(i)$ is a Gaussian integer that can a priori vanish by cancellation.

We henceforth treat the even case $n = 2m$ and write $P_\lambda := \langle s_\lambda, \psi^m \rangle = G_\lambda(i) \in \mathbb{Z}[i]$. The odd case is recovered from $G_\lambda(i) = \langle s_\lambda, h_1 \bar{\psi}^m \rangle = \sum_\mu \overline{P_\mu}$, the sum over the even partitions μ obtained by deleting one corner box of λ .

3 Three faces of P_λ

3.1 The interpolating polynomial and the skewing recursion

Let $P_\lambda(t) := \langle s_\lambda, (h_2 + t e_2)^m \rangle \in \mathbb{Z}_{\geq 0}[t]$, so $P_\lambda = P_\lambda(i)$. Because $\langle s_\lambda, (h_2 + t e_2)^m \rangle$ is a product of the Schur-positive functions h_2, e_2 , its coefficients are nonnegative integers. The skewing operators $h_2^\perp, e_2^\perp$ remove horizontal, resp. vertical, 2-strips, giving the recursion

$$P_\lambda(t) = \sum_{\lambda/\mu \text{ horiz. 2-strip}} P_\mu(t) + t \sum_{\lambda/\mu \text{ vert. 2-strip}} P_\mu(t), \quad P_\emptyset = 1. \quad (1)$$

Three evaluations are pinned down: $P_\lambda(1) = \langle s_\lambda, p_1^{2m} \rangle = f^\lambda$; $P_\lambda(-1) = \langle s_\lambda, p_2^m \rangle = \chi^\lambda(2^m) = \pm \binom{m}{|\lambda^0|} f^{\lambda^0} f^{\lambda^1}$ (the signed number of standard domino tableaux, nonzero iff λ has empty 2-core); and $P_\lambda(0) = K_{\lambda, (2^m)}$. Conjugation acts by coefficient reversal,

$$P_{\lambda'}(t) = t^m P_\lambda(1/t), \quad \text{hence} \quad P_{\lambda'}(i) = i^m \overline{P_\lambda(i)}, \quad (2)$$

using $\omega(h_2 + t e_2) = t(h_2 + t^{-1} e_2)$. In particular $G_\lambda(i) = 0 \iff G_{\lambda'}(i) = 0$, and the self-conjugate $(2, 2)$ is consistent with this.

3.2 Hermite determinant and Gaussian integral

Setting $p_{\geq 3} = 0$ is the ring homomorphism $\Lambda \rightarrow \mathbb{C}[A, B]$, $p_1 \mapsto A$, $p_2 \mapsto B$. Under it $h_r \mapsto [z^r] \exp(Az + \frac{B}{2}z^2)$, an Appell/Hermite polynomial, so by Jacobi–Trudi $s_\lambda(p_1=A, p_2=B, 0) = \det(h_{\lambda_i-i+j}(A, B, 0))$ is a Hermite determinant. A Hubbard–Stratonovich linearisation of p_1^2 then yields the integral representation

$$P_\lambda(t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} s_\lambda(p_1=2s, p_2=2t, p_{\geq 3}=0) ds, \quad (3)$$

equivalently $G_\lambda(i) = \mathbb{E}_{Y \sim N(0,2)} [s_\lambda(p_1=Y, p_2=-2i, 0)]$ up to the unit $\beta^m i^m$, $\beta = \frac{1-i}{2}$. (Verified numerically for all λ with $n \leq 10$.)

3.3 A Parseval identity

By orthonormality of Schur functions,

$$\sum_{\lambda \vdash 2m} |G_\lambda(i)|^2 = \langle \psi^m, \bar{\psi}^m \rangle = 2^{-m} \sum_{a=0}^m \binom{m}{a}^2 2^a a! (2m-2a)! > 0, \quad (4)$$

since $\langle p_\mu, p_\nu \rangle = z_\mu \delta_{\mu\nu}$ and $\alpha\beta = \frac{1}{2}$ for $\psi = \alpha p_1^2 + \beta p_2$ ($\alpha = \frac{1+i}{2}$). (Verified for $m \leq 5$: the totals are 2, 12, 168, 4512, 198720.) This shows the fiber values are not all zero but does not by itself isolate $(2, 2)$.

4 What is proved, and the boundary

Proposition 2 (Extreme shapes). $P_{(2m)}(i) = 1$ and $P_{(1^{2m})}(i) = i^m$; both are nonzero.

Proof. A one-row shape admits only horizontal 2-strip removals, and there is a unique chain $(2m) \rightarrow (2m-2) \rightarrow \cdots \rightarrow \emptyset$; by (1) its weight is t^0 , so $P_{(2m)}(t) = 1$. Dually $P_{(1^{2m})}(t) = t^m$ by (2). $\square$

Proposition 3 (Kostka expansion and a Gaussian-prime obstruction). Since $e_2 = h_1^2 - h_2$, one has $\psi = i h_1^2 + (1-i)h_2$, hence

$$\psi^m = \sum_{k=0}^m \binom{m}{k} i^{m-k} (1-i)^k h_{(2^k 1^{2m-2k})}, \quad \text{so} \quad G_\lambda(i) = i^m \sum_{k=0}^m \binom{m}{k} (-1-i)^k K_{\lambda, (2^k 1^{2m-2k})}.$$

Let $(1+i)$ be the Gaussian prime above 2. Reducing modulo $(1+i)$ kills every $k \geq 1$ term, leaving $G_\lambda(i) \equiv i^m K_{\lambda, (1^{2m})} = i^m f^\lambda \pmod{(1+i)}$. Therefore

$$f^\lambda \text{ odd} \implies G_\lambda(i) \neq 0.$$

In particular every vanisher has f^λ even; consistent with $f^{(2,2)} = 2$. (Both displays verified for all λ with $n \leq 12$, 0 exceptions; the sufficient condition has no exception through $n = 14$.)

Proof. The expansion is the binomial theorem applied to $\psi = i h_1^2 + (1-i)h_2$ together with $h_1^{2(m-k)} h_2^k = h_{(2^k 1^{2m-2k})}$; pairing with s_λ uses $\langle s_\lambda, h_\mu \rangle = K_{\lambda\mu}$ and $i^{m-k} (1-i)^k = i^m ((1-i)/i)^k = i^m (-1-i)^k$. Since $(-1-i) = -(1+i)$, each $k \geq 1$ summand lies in $(1+i)\mathbb{Z}[i]$, so $G_\lambda(i) \equiv i^m f^\lambda \pmod{(1+i)}$. As $(1+i) \mid f^\lambda$ in $\mathbb{Z}[i]$ iff $2 \mid f^\lambda$, an odd f^λ forces $G_\lambda(i) \not\equiv 0$, hence $G_\lambda(i) \neq 0$. $\square$

Remark 2. The obstruction is sharp at the level of one prime: it cannot see vanishers, which all have f^λ even. One might hope to iterate $(1+i)$ -adically, but the even- and odd- k partial sums do *not* separately vanish at a vanisher (for $(2, 2)$ they are $2 + 2i$ and $-2 - 2i$, which cancel), so a single congruence does not suffice; controlling $v_{(1+i)}(G_\lambda(i))$ in general appears to require the 2-adic valuations of the Kostka numbers $K_{\lambda, (2^k 1^{2m-2k})}$ (the 2-core tower).

The single congruence of Proposition 3 refines, at no extra cost, into a statement about the real and imaginary parts separately, which sharpens the necessary conditions for a vanisher.

Proposition 4 (Refined parity congruence). *Write $G_\lambda(i) = a + bi$ and $\chi := \chi^\lambda(2^m)$. Then*

$$a = \operatorname{Re} G_\lambda(i) \equiv \frac{1}{2}(f^\lambda + \chi), \quad b = \operatorname{Im} G_\lambda(i) \equiv \frac{1}{2}(f^\lambda - \chi) \pmod{2}.$$

Proof. By (1), $G_\lambda(i) = \sum_{v \geq 0} a_v i^v$ with $a_v \in \mathbb{Z}_{\geq 0}$. Grouping by $v \bmod 4$ and writing $n_r = \sum_{v \equiv r(4)} a_v$, we have $a = n_0 - n_2 \equiv n_0 + n_2 \pmod{2}$ and $b = n_1 - n_3 \equiv n_1 + n_3 \pmod{2}$. Now $n_0 + n_2 = \sum_{v \text{ even}} a_v =: N_e$ and $n_1 + n_3 = \sum_{v \text{ odd}} a_v =: N_o$ satisfy $N_e + N_o = P_\lambda(1) = f^\lambda$ and $N_e - N_o = P_\lambda(-1) = \chi$, so $N_e = \frac{1}{2}(f^\lambda + \chi)$, $N_o = \frac{1}{2}(f^\lambda - \chi)$. $\square$

Corollary 1 (Mod-4 necessary conditions). *If $G_\lambda(i) = 0$ then f^λ and $\chi^\lambda(2^m)$ are even and $f^\lambda \pm \chi^\lambda(2^m) \equiv 0 \pmod{4}$. In particular, if λ has nonempty 2-core then $\chi^\lambda(2^m) = 0$ and one needs $4 \mid f^\lambda$; if λ has empty 2-core then, by the Stanton–White formula [Clio 2026-06-02, Thm. 1], $\chi^\lambda(2^m) = (-1)^{n(\lambda)} f^{\lambda^0} f^{\lambda^1} \binom{m}{|\lambda^0|}$, and one needs $f^\lambda \pm \chi^\lambda(2^m) \equiv 0 \pmod{4}$.*

Proof. $G_\lambda(i) = 0$ forces $a = b = 0$, so both $\frac{1}{2}(f^\lambda \pm \chi)$ are even by Proposition 4; hence $f^\lambda \pm \chi \equiv 0 \pmod{4}$, forcing f^λ, χ even. The class (2^m) has no fixed points, so $\chi^\lambda(2^m) \neq 0$ only when λ is tileable by m dominoes, i.e. has empty 2-core; the empty-core value is the cited product ($c_0 = 0$). $\square$

These congruences exclude infinitely many shapes (all odd- f^λ , and more), but infinitely many shapes still satisfy them, so the parity criterion alone cannot terminate.

Proposition 5 (The $m = 2$ support, exactly). $\psi^2 = s_4 + (1 + 2i)s_{31} + (2i - 1)s_{211} - s_{14}$. Hence $\operatorname{supp}_{\mathbb{C}}(\psi^2) = \{(4), (3, 1), (2, 1, 1), (1^4)\} = \{\lambda \vdash 4\} \setminus \{(2, 2)\}$, and $(2, 2)$ is the unique partition of 4 with $G_\lambda(i) = 0$.

Proof. With $h_2 = s_2$, $e_2 = s_{11}$ and Pieri, $s_2^2 = s_4 + s_{31} + s_{22}$, $s_{11}^2 = s_{22} + s_{211} + s_{14}$, $s_2 s_{11} = s_{31} + s_{211}$. Therefore $\psi^2 = (s_2 + i s_{11})^2 = s_2^2 + 2i s_2 s_{11} - s_{11}^2 = s_4 + (1 + 2i)s_{31} + (1 - 1)s_{22} + (2i - 1)s_{211} - s_{14}$. The coefficient of s_{22} is $1 - 1 = 0$: the real parts coming from s_2^2 and $-s_{11}^2$ cancel, and the imaginary cross term $2i s_2 s_{11}$ contributes nothing to s_{22} . Equivalently $G_{(2,2)}(i) = \chi^{(2,2)}(1^4)\alpha^2 + \chi^{(2,2)}(2, 2)\beta^2 = 2(\alpha^2 + \beta^2) = 2 \cdot 0 = 0$ since $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 - 1 = 0$. $\square$

4.1 Exact values on hooks and two rows

The skewing recursion (1) stays inside the hook and two-row families and solves there in closed form. We use the weights of a 2-strip removal λ/μ read off from (1): 1 if λ/μ is a connected horizontal domino, i if connected vertical, and $1 + i$ if λ/μ is a disconnected pair (distinct row and column), the last because such a pair is simultaneously a horizontal and a vertical 2-strip.

Lemma 2 (Near-rectangular hook). *For $M \geq 1$ ($n = 2M$), $G_{(2M-1,1)}(i) = (M - 1) + M i$, which is never zero.*

Proof. From $(a, 1)$ with $a = 2M - 1$ the removable 2-strips are: the arm pair $(1, a), (1, a - 1)$ (horizontal, $\rightarrow (a - 2, 1)$, weight 1, present iff $a \geq 3$), and the disconnected pair $(1, a), (2, 1)$ (distinct row and column as $a \geq 2$, $\rightarrow (a - 1)$, weight $1 + i$). With $G_{(a-1)} = 1$ (Prop. 2) this gives, for $M \geq 2$, $G_{(2M-1,1)} = G_{(2M-3,1)} + (1 + i)$. The base $G_{(1,1)} = i$ (a single vertical domino) solves the recursion to $i + (M - 1)(1 + i) = (M - 1) + Mi$. $\square$

Theorem 2 (The two-row family $(2m - 2, 2)$). *For $m \geq 2$ ($n = 2m$),*

$$\boxed{G_{(2m-2,2)}(i) = m(m-2)i},$$

which vanishes if and only if $m = 2$, i.e. if and only if $\lambda = (2, 2)$.

Proof. Put $a = 2m - 2$. For $a \geq 4$ the removable 2-strips of $(a, 2)$ are exactly: the arm pair $(1, a), (1, a - 1)$ (horizontal, $\rightarrow (a - 2, 2)$, weight 1); the second-row pair $(2, 1), (2, 2)$ (horizontal, $\rightarrow (a)$, weight 1); and the disconnected pair $(1, a), (2, 2)$ (distinct row and column since $a > 2$, $\rightarrow (a - 1, 1)$, weight $1 + i$). These exhaust the partition-preserving deletions of two cells, the only corners being $(1, a)$ and $(2, 2)$. Hence, with $g_m := G_{(2m-2,2)}(i)$ and using $G_{(a)} = 1$ (Prop. 2) and $G_{(a-1,1)} = (m - 2) + (m - 1)i$ (Lemma 2, $M = m - 1$), for $m \geq 3$:

$$g_m = g_{m-1} + 1 + (1 + i)[(m - 2) + (m - 1)i] = g_{m-1} + 1 + (-1 + (2m - 3)i) = g_{m-1} + (2m - 3)i,$$

since $(1 + i)[(m - 2) + (m - 1)i] = (m - 2) - (m - 1) + [(m - 1) + (m - 2)]i = -1 + (2m - 3)i$. The base is $g_2 = G_{(2,2)}(i) = 0$ (Proposition 5). Summing, $g_m = i \sum_{k=3}^m (2k - 3) = i[(m - 1)^2 - 1] = m(m - 2)i$, zero iff $m = 2$. $\square$

Corollary 2. *Among two-row shapes (a, b) the value $G_{(a,b)}(i)$ vanishes only at $(2, 2)$, for $b \leq 2$ (rows, $b=0$; near-rectangular, $b=1$; Theorem 2, $b=2$); the cases $b \geq 3$ are verified vanishing-free for $n \leq 24$.*

Theorem 3 (Computational verification). *For every λ with $|\lambda| \leq 16$ (both parities; 913 shapes through $n = 16$), $G_\lambda(i) = 0$ if and only if $\lambda = (2, 2)$; rechecked through $n = 18$.*

5 The remaining statement and why it is hard

By Theorem 1 the full theorem $G_\lambda(i) = 0 \iff \lambda = (2, 2)$ is *equivalent* to the following.

Conjecture 1 (Full support). *For $m \geq 3$, $\text{supp}_{\mathbb{C}}(\psi^m) = \{\lambda \vdash 2m\}$; and for $m = 2$ the support omits exactly $(2, 2)$. Equivalently, $P_\lambda(i) \neq 0$ for all $\lambda \neq (2, 2)$.*

The $m = 2$ clause is Proposition 5; the $m = 1$ case is $\psi = s_2 + is_{11}$ (full). What remains is the $m \geq 3$ clause.

Why positivity fails. Writing $P_\lambda(i) = \text{Re } P_\lambda(i) + i \text{Im } P_\lambda(i)$, the real and imaginary parts are the signed sums $\text{Re} = \sum_k (-1)^k a_{2k}$, $\text{Im} = \sum_k (-1)^k a_{2k+1}$ where $P_\lambda(t) = \sum_j a_j t^j$, $a_j \geq 0$. Neither is sign-definite, and neither vanishing-set is empty: e.g. at $m = 3$, $\text{Re } P_{(4,2)}(i) = 0$ while $\text{Im } P_{(2,2,1,1)}(i) = 0$. The content of Conjecture 1 is that the two vanishing-sets are *disjoint* (except at $(2, 2)$). There is no single Schur-positive function whose support computes $P_\lambda(i)$, because ψ^m is generically irreducible as a t -polynomial $P_\lambda(t)$ (for instance $P_{(5,5)}(t) = t^5 + 10t^3 + 10t^2 + 15t + 6$).

Two routes, and where each stalls. The recursion (1) at $t = i$ reads $P_\lambda(i) = \sum_{\text{horiz}} P_\mu(i) + i \sum_{\text{vert}} P_\mu(i)$, a sum of Gaussian integers; the model $P_\lambda(i) = \sum_{\pi} i^{\text{vert}(\pi)}$ runs over chains removing 2-strips. At $t = -1$ the analogous signed sum $\sum_{\pi} (-1)^{\text{vert}(\pi)} = \chi^\lambda(2^m)$ is resolved by a classical sign-reversing involution (domino tableaux), giving $\text{ord}_{q=-1} G_\lambda = \lfloor |2\text{-core}|/2 \rfloor$. The tempting ζ_4 analogue – a sign-reversing 2×2 -flip involution whose fixed set is 4-quotient-indexed – *fails*: it is not an involution (overlapping flip windows), and the value $\sum i^{\text{vert}}$ over its non-fixed chains is a nonzero set-invariant (first breaking at $(3, 3)$, where the non-fixed chains form an unpairable 3-cycle), so no such involution can exist. Two routes remain.

1. **A $(1+i)$ -adic monovariant.** It suffices that $v_{1+i}(G_\lambda(i))$ be finite for $\lambda \neq (2, 2)$. Computation confirms finiteness but shows the valuation is *irregular* (values $2, 3, 5, 6, 8, 9, 11, \dots$, no closed form against the 2-core/4-quotient data), so the clean $q = -1$ law does not persist at i . One would need a combinatorial valuation-minimal, uniquely-placed predecessor in (1); I could not pin one uniformly. (The staircase 4-core $(4, 3, 2, 1)$ has $G(i) = 24(1+i)^5$, a pure prime power, $v_{1+i} = 11$.)
2. **A Hermitian lower bound.** From $\psi^m \bar{\psi}^m = \Xi^m$ with $\Xi = \psi \bar{\psi} = h_2^2 + e_2^2 = s_4 + s_{31} + 2s_{22} + s_{211} + s_{14}$ (Schur-positive, *full* support of partitions of 4, unlike the $d = 3$ Θ), one has for every $\rho \vdash 4m$

$$0 \leq \langle s_\rho, \Xi^m \rangle = \sum_{\lambda, \mu \vdash 2m} c_{\lambda\mu}^\rho G_\lambda(i) \overline{G_\mu(i)},$$

a nonnegative Hermitian form. A ρ with a *unique* Littlewood–Richardson halving $\rho = \lambda + \lambda$ would give $0 < \langle s_\rho, \Xi^m \rangle = c_{\lambda\lambda}^\rho |G_\lambda(i)|^2$ and finish; but no such “primitively halvable” ρ exists in general (e.g. $\rho = 2\lambda$ has several halvings), so the off-diagonal terms cannot be discarded.

The natural home for the missing positivity is the even (pair-of-boxes) Fock-space formalism of the seed: $\psi^m = \left(\frac{1+i}{2}\right)^m (p_1^2 - i p_2)^m$ is a genuinely complex tau-type vector, not a positive one.

6 Computational verification

All identities were checked by exact computation (Gaussian integers in $\mathbb{Z}[i]$, Schur expansions via Pieri; scripts in `~/projects/scratch/2026-06-09-d4-involution/`). Specifically: the reduction Theorem 1 and $G_\lambda(i) = P_\lambda(i)$ against direct evaluation, $n \leq 14$ (0 mismatches); the integral representation (3) and Parseval identity (4) for $m \leq 5$; the conjugation symmetry (2); the basic and refined parity congruences (Prop. 3, Prop. 4) for all λ , $n \leq 16$; Propositions 2, 5; the closed forms $G_{(2M-1,1)} = (M-1) + Mi$ and $G_{(2m-2,2)} = m(m-2)i$ (Lemma 2, Theorem 2) for $m \leq 9$; two-row shapes vanishing-free except $(2, 2)$ for $n \leq 24$ and hooks for $n \leq 22$ (Cor. 2); and the master scan (Theorem 3, $n \leq 16$, rechecked $n \leq 18$). The non-single-valuedness of ψ^m through the 2-quotient, and the irregularity of $v_{1+i}(G_\lambda(i))$, were checked for $n \leq 14$.